

Topic: Entity-Relationship Diagram & Database Design for International Airport System

Diagram Legend

Symbol / Notation	Name	Description
PK	Primary Key	Unique identifier for each record in an entity table.
FK	Foreign Key	Attributes referencing the Primary Key of another table to establish relationships.
1 : N	One-to-Many	One record in the parent table relates to multiple records in the child table.
1 : 1	One-to-One	One record in the parent table relates to exactly one record in the child table.

3. Entities, Attributes, and Data Types

The system consists of 10 primary entities, each structured with explicit primary/foreign keys and data types:

- **Airport:** Stores physical airport details (airport_id: INT [PK], airport_name: VARCHAR, country: VARCHAR, state: VARCHAR, city: VARCHAR, created_at: TIMESTAMP, updated_at: TIMESTAMP).
- **Airline:** Stores airline company profiles (airline_id: INT [PK], airline_code: VARCHAR, name: VARCHAR, country: VARCHAR, created_at: TIMESTAMP, updated_at: TIMESTAMP).
- **Flight:** Tracks scheduled and actual flights, gate assignments, and aircraft linkage (flight_id: INT [PK], airline_id: INT [FK], departure_airport_id: INT [FK], arrival_airport_id: INT [FK], departing_gate: VARCHAR, arriving_gate: VARCHAR, scheduled_departure_time: TIMESTAMP, scheduled_arrival_time: TIMESTAMP, actual_departure_time: TIMESTAMP, actual_arrival_time: TIMESTAMP, created_at: TIMESTAMP, updated_at: TIMESTAMP).
- **Passenger:** Stores passenger accounts, identity data, and frequent flyer records (passenger_id: INT [PK], first_name: VARCHAR, last_name: VARCHAR, gender: VARCHAR, date_of_birth: DATE, country_of_citizenship: VARCHAR, country_of_residence: VARCHAR, passport_number: VARCHAR, created_at: TIMESTAMP, updated_at: TIMESTAMP).
- **Booking:** Links passenger accounts to scheduled flight legs (booking_id: INT [PK], flight_id: INT [FK], passenger_id: INT [FK], status: VARCHAR, booking_platform: VARCHAR, ticket_price: DECIMAL, created_at: TIMESTAMP, updated_at: TIMESTAMP).
- **Booking_Change_History:** Audit log tracking flight ticket modifications (change_id: INT [PK], booking_id: INT [FK], change_details: TEXT, changed_at: TIMESTAMP).
- **Boarding_Pass:** Issued boarding documents for individual bookings (boarding_pass_id: INT [PK], booking_id: INT [FK], seat: VARCHAR, boarding_time: TIMESTAMP, created_at: TIMESTAMP, updated_at: TIMESTAMP).

TIMESTAMP).

- **Baggage:** Tracks checked luggage items associated with bookings (baggage_id: INT [PK], booking_id: INT [FK], weight_kg: DECIMAL, created_at: TIMESTAMP, updated_at: TIMESTAMP).
- **Baggage_Check:** Records inspection results for registered luggage (baggage_check_id: INT [PK], booking_id: INT [FK], passenger_id: INT [FK], check_results: VARCHAR, created_at: TIMESTAMP, updated_at: TIMESTAMP).
- **Security_Check:** Logs passenger security screening results (security_check_id: INT [PK], passenger_id: INT [FK], check_results: VARCHAR, created_at: TIMESTAMP, updated_at: TIMESTAMP).

4. Relationships and Constraints

This database facilitates the management of airport-related operations, including passenger bookings, flight scheduling, aircraft information, and frequent flyer programs, ensuring efficient and organized airport services. These tables establish various types of relationships, including:

- **One-to-Many (1:N):** One record in one table corresponds to many records in another table (e.g., Airline to Flight, Passenger to Booking).
- **Many-to-One (N:1):** Many records in one table correspond to one record in another table (e.g., multiple Flight records link to one departure Airport).
- **One-to-One (1:1):** Each record in one table is related to exactly one record in another table, and vice versa (e.g., Booking to Boarding_Pass).
- **Many-to-Many (N:M):** Many records in one table can be related to many records in another table through intermediate linking tables (e.g., Passenger and Flight through Booking).